8 Comparing self-serve data platforms

This chapter covers

- Platforms leveraging Google Cloud Platform
- Platforms leveraging Amazon Web Services
- Platforms leveraging Databricks
- Platforms leveraging Apache Kafka
- Pros and cons of each platform

In the first part of this book, you learned all the basics of the data mesh. In chapter 7, you did a deep dive into building self-serve platforms using the example of Messflix. You learned how to iterate on one specific example and build up a fully functioning platform.

This chapter is different. While the data mesh paradigm is not tied to any one implementation, we believe it's necessary to see it in action more than once to really find a suitable architecture for your platform.

This is the focus of this chapter: supplying you with multiple architectures and technology stacks to make it easy for you to choose a good architecture. We will walk you through four possible architectures that all are inspired by real-world examples implemented at multiple companies throughout the world.

NOTE Each section highlights a specific choice of cloud provider, data catalog, and programming language to make these rather complex architectures easy to understand. However, all architectures can be mixed and matched or translated to different technology combinations. The architecture in section 8.1 on Google Cloud Platform, for instance, could be translated almost one-to-one to Amazon Web Services. But also keep in mind that building the technical side of data meshes is unusually hard compared to other technology projects. From our perspective, the reason is that we simply don't yet have many data mesh-specific tools, so we are using and reusing a lot of basic tools and need to be mindful of their exact usage. You will experience that directly in the first section, where a minor detail like service accounts becomes essential for the data mesh.